NOT Gate Simulation

Implementation and verification method

1. Objective

Implement a one-bit NOT gate in VHDL and verify its behavior using Logisim-evolution with the Questa VHDL simulator.

2. VHDL source: not_gate.vhd

```
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
    port (
        a : in  std_logic;
        y : out std_logic
    );
end not_gate;

architecture rtl of not_gate is
begin
    y <= not a;
end rtl;
```

3. Circuit and procedure

The circuit connects input pin a to the not_gate VHDL entity and connects its output y to an output pin. Both screenshots on page 2 show the circuit. VHDL simulation was enabled in Logisim-evolution. The input was toggled with the Poke tool, and the output was checked for both binary inputs.

4. Expected and observed results

Input a	Expected y	Observed y	Result
0	1	1	Pass
1	0	0	Pass

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Circuit screenshots and discussion

5. Logisim simulation screenshots

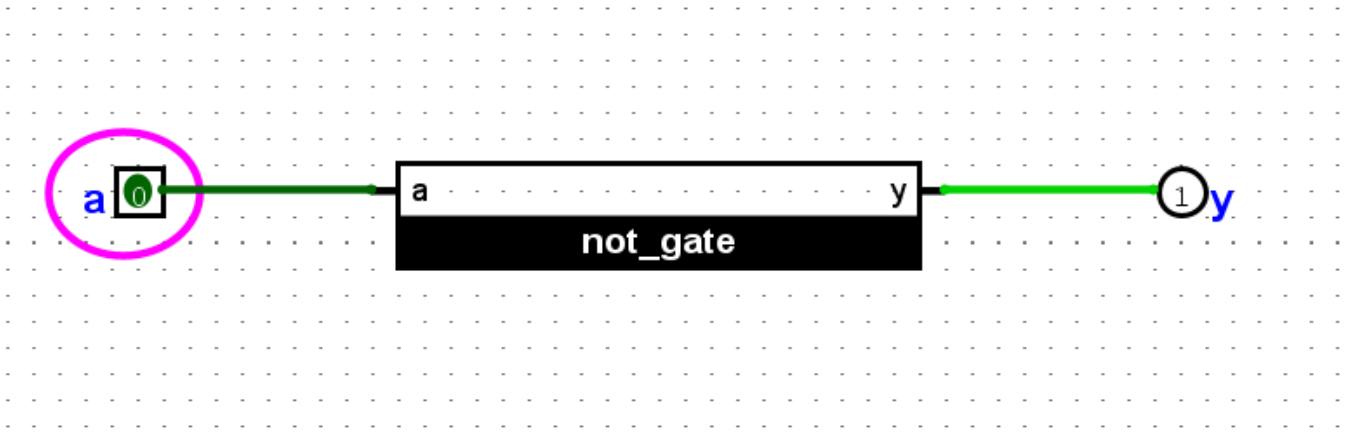

Figure 1. Input a = 0; output y = 1.

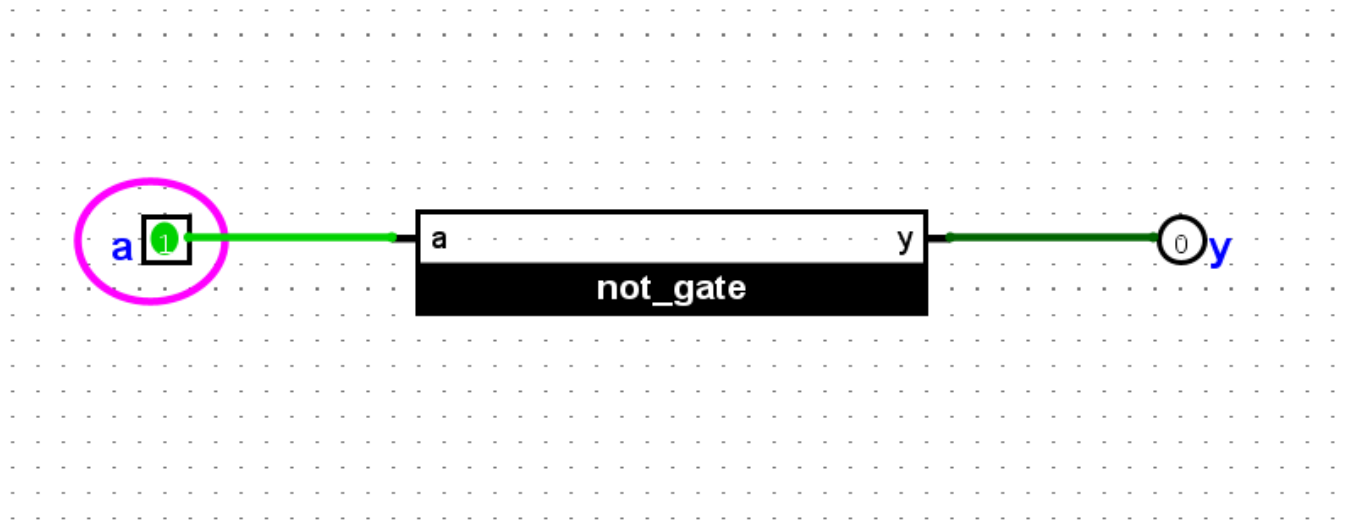

Figure 2. Input a = 1; output y = 0.

6. Discussion and conclusion

The statement y <= not a assigns the logical inverse of a to y.
The screenshots show y = 1 when a = 0 and y = 0 when a = 1.
Both observed results match the NOT-gate truth table, confirming correct binary inversion in the Logisim and Questa simulation workflow.